

Amitej Singh Datta

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Professional Summary

AI/ML focused Computer Science graduate currently pursuing an M.S. in Data Science & Machine Learning at NUS, specializing in computer vision and medical image segmentation. Published first-author work (ICCV 2023, Springer) on a hybrid deep learning architecture for prostate segmentation, and have separately developed parameter-efficient fine-tuning approaches for foundation vision models that matched state-of-the-art accuracy while significantly cutting training time. Proficient in PyTorch, TensorFlow, OpenCV, and Hugging Face across the full model lifecycle. Known for methodical experimentation, clear technical communication, and following problems through to a working, well-benchmarked solution.

Core Competencies

- **AI & Machine Learning:** Deep Learning, CNNs, Transformers, Clustering, Model Training & Evaluation
- **Computer Vision:** Image Segmentation, Object/Pose Detection, YOLO, ViT, OpenCV, MediaPipe
- **Foundation Models:** LoRA/PEFT, RAG, Hugging Face, MedSAM
- **ML Engineering:** PyTorch, TensorFlow, scikit-learn, NumPy, FastAPI, Flask
- **Software Development:** Python, C++, Java, JavaScript, SQL, React, Git, Linux
- **Research:** Experimentation, Model Benchmarking, Hyperparameter Tuning, Technical Communication
- **Professional Strengths:** Analytical problem-solving, attention to detail, collaboration, continuous learning

Education

Vellore Institute of Technology, Chennai B.Tech, Computer Science & Engineering (AI & ML)	Sep 2022 – July 2026 GPA: 9.09
National University of Singapore M.S., Data Science & Machine Learning	Aug 2026 – Present

Experience

Software Intern · Casamed HealthTech (Remote)	Oct 2024 – Jan 2025
• Automated extraction and analysis of key metrics from medical reports using Python, streamlining clinical data pipelines and reducing manual processing overhead across the platform. • Developed pose-detection scripts (OpenCV, MediaPipe) to enable AI-assisted physiotherapy assessments; contributed backend API improvements to enhance ML model integration and data flow.	
Student Trainee Intern · DRDO – CAIR, Bengaluru	May 2025 – Jul 2025
• Built a video summarisation pipeline using ResNet-50 for frame-level feature extraction; applied unsupervised clustering (K-Means, DBSCAN, motion-based segmentation) to identify keyframes and generate compact summaries. • Evaluated system against TVSum benchmark annotations (Precision, Recall, F1-score); deployed end-to-end via a Flask-React web application supporting video upload, processing, and summary playback.	

Projects

AI-based Prostate Segmentation on Micro-Ultrasound Images — <i>Published, ICCV 2023</i>	github/ProstateSegmentation
• Designed a hybrid VGG16-UNet architecture to automate prostate volume delineation on micro-ultrasound scans, replacing a labour-intensive manual workflow and reducing radiologist annotation time. • Applied data augmentation and hyperparameter tuning to improve model generalisation; implemented post-processing boundary refinement to ensure precise segmentation outputs for clinical use. • Presented findings at ICCV 2023 (BITS Goa); paper accepted for publication in the Springer Book Series, validating clinical relevance and technical rigour of the approach.	
LoRA Fine-tuning Framework for Medical Image Segmentation	github/MedSAM_LoRA
• Implemented a from-scratch LoRA (Low-Rank Adaptation) trainer and applied it to MedSAM (ViT-B) for prostate segmentation, achieving 95.7% Dice / 91.9% IoU — surpassing the VGG-UNet baseline with 82% fewer trainable parameters. • Optimised training via image embedding caching, reducing per-epoch time from 60 minutes to 3 minutes (60% overall speedup) while maintaining state-of-the-art segmentation performance.	
Low-Cost Virtual Reality System Using Smartphone and Webcam	github/VisioLink
• Engineered a budget VR system using a smartphone (Google Cardboard) as the display and a laptop webcam for head and hand motion tracking via OpenCV and MediaPipe, eliminating costly hardware dependencies. • Implemented WebSocket-based bidirectional communication between laptop and phone, achieving low-latency real-time interaction comparable to OpenTrack-style solutions without Oculus controllers or full-body trackers. • Designed the system to be hardware-agnostic, enabling deployment on any Android device and standard webcam — making immersive VR accessible without specialist equipment.	

Extracurricular

- **Culture Lead**, Fraternity of Young Innovators, VIT Chennai
- **E-Summit VIP GuestCare Head**, Voyage (college tech summit)
- **Volunteer**, Khalsa Aid (Punjab flood relief operations) and SOS Children Villages of India.